

INFORMATION REQUIREMENTS ANALYSIS

MODULE 3

LEARNING OBJECTIVES

What you will learn

ILO 4: Elicit, analyze, prioritize, specify, validate, and trace stakeholder and system requirements using ethical and inclusive methods.

CLO alignment: Build trustworthy requirements evidence for CLO 2 and CLO 3, then carry it into the progressive Title Proposal portfolio.

Module 3 starts a new diagnostic cycle

DA 2 must be completed in Canvas before instruction; it maps readiness for Modules 3–4.



Diagnostic results are ungraded. They guide adaptive practice and targeted support—not course grades.

Question of understanding

How do you know a requirement represents a real stakeholder need—not an analyst's assumption?



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STAKEHOLDER AND REQUIREMENTS EVIDENCE

Requirements analysis is an evidence discipline

The analyst converts stakeholder experience and organizational records into defensible system needs.



A persuasive statement is not automatically a verified requirement.

Stakeholders are defined by impact and knowledge

Identify people who use, operate, approve, support, regulate, supply data to, or are affected by the system.



Do not limit discovery to the loudest, easiest, or highest-ranking participants.

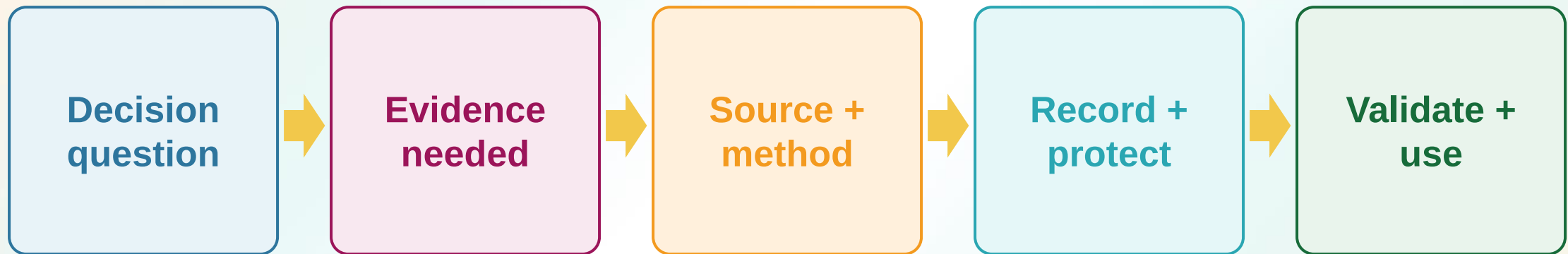
Inclusive discovery makes missing perspectives visible

Review stakeholder power, process knowledge, exposure to change, access barriers, and potential harm.

Lens	Ask	Risk if omitted
Decision power	Who can authorize or block?	Unapproved requirements
Process knowledge	Who handles real exceptions?	Incomplete workflow
Change impact	Whose work or rights change?	Adoption or harm
Access + inclusion	Who may be underrepresented?	Biased design

Plan evidence before choosing a method

Start with the decision question, then select credible sources, methods, records, and validation steps.



Method choice should follow the evidence gap—not personal convenience.

Use interactive and unobtrusive methods together

Interactive methods explain meaning; unobtrusive methods reveal actual records, patterns, and conditions.

INTERACTIVE

Interviews • workshops/JAD
questionnaires • facilitated sessions

Best for goals, meanings, exceptions,
conflict

UNOBTRUSIVE

Observation • documents • policies
records • logs • analytics • scanning

Best for behavior, volume, sequence,
and constraints

Triangulation reduces dependence on any one person or
source.



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Interviews uncover goals, exceptions, and reasoning

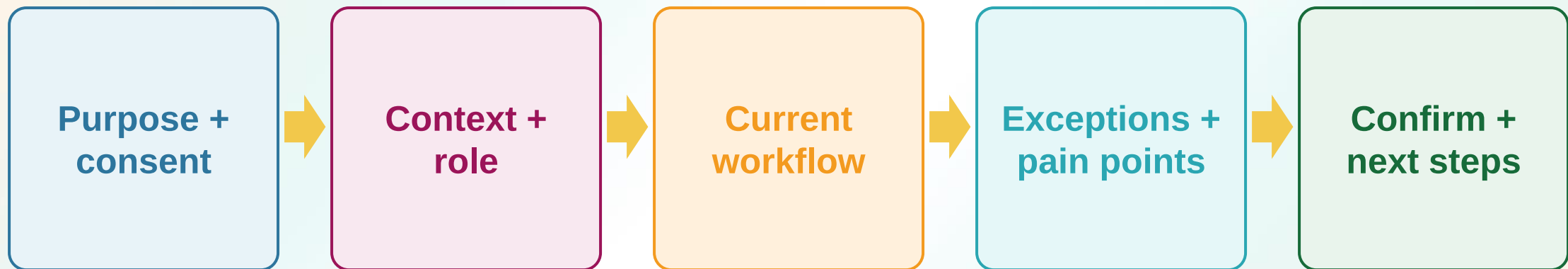
Use interviews when depth matters and the analyst needs to follow the participant's experience.

Strong use	Evidence sought	Watch for
Process owner	Rules + decisions	Official view only
Frontline user	Workarounds + exceptions	Single-user bias
Support role	Failures + recovery	Technical jargon
Affected person	Access + impact	Power imbalance



A strong interview protocol protects focus and trust

Prepare a consistent path while allowing relevant follow-up questions.



Record date, participant role, version, approved notes, confidentiality level, and unresolved questions.



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Good questions invite evidence instead of agreement

Move from open questions to focused probes; avoid leading, double-barreled, and solution-first wording.

WEAK

“Wouldn’t an AI dashboard make approvals faster and more accurate?”

STRONG

“Walk me through the last approval. Where did delay occur, and what evidence was unavailable?”

Useful probes: What happened next? How often? Show me an example. Who decides? What changes in an exception?



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Workshops and JAD make differences discussable

Bring the right roles together to map the process, surface conflicts, agree terminology, and record decisions.



A workshop creates evidence only when contributions, decisions, dissent, and follow-ups are documented.



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Workshop controls protect evidence quality

Assign roles, time-box decisions, separate facts from proposals, and record disagreement.

Role / control	Responsibility	Evidence produced
Facilitator	Neutral process + participation	Agenda + discussion record
Participants	Domain evidence + review	Examples + decisions
Recorder	Trace claims and changes	Decision / issue log
Approver	Confirm agreed baseline	Approval or conditions



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Questionnaires scale comparable questions

Use them when many respondents can answer the same clearly defined questions and coverage matters more than depth.

GOOD FIT

Frequency • preference • satisfaction
known categories • broad coverage

Use follow-up interviews for
explanation

POOR FIT

Unfamiliar process • complex
exceptions
ambiguous terminology • sensitive
issues

A form cannot repair a weak question



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Build clear, answerable questionnaires

Pilot each item before release and connect it to a defined analysis decision.

Quality check	Weak item	Better design
Single idea	Fast and secure?	Separate speed and security
Neutral wording	How helpful is our excellent portal?	How helpful is the portal?
Complete options	Daily / weekly / never	Add monthly + other
Recall period	Usually	In the last 30 days

Observation reveals the work people actually perform

Define the setting, task, events, timing, handoffs, artifacts, exceptions, and observer boundaries.



Do not record sensitive screens, credentials, identities, or conversations beyond the approved purpose.

Documents reveal formal rules and expected evidence

Review forms, policies, reports, contracts, approval records, and data definitions.

Inspect	Learn	Verify
Forms + templates	Required fields + handoffs	Are fields used?
Policies + rules	Authority + constraints	Is practice consistent?
Reports + records	Outputs + decisions	Who relies on them?
Definitions	Meaning + ownership	Are terms consistent?

Records and logs reveal patterns people may not recall

Use approved samples to test volume, timing, frequency, error, exception, and data-quality claims.

Source	Possible evidence	Safeguard
Transaction records	Volume + turnaround	Minimize fields
System logs	Events + failures	Protect identifiers
Analytics	Patterns + drop-offs	Check definitions
External scan	Technology + policy change	Verify authority



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Ethical discovery protects people and project credibility

Collect only necessary information and state how it will be protected.

Consent

**Purpose +
choice**

Confidential

**Controlled
access**

Minimization

**Only needed
data**

De-identify

**Remove direct
identity**

Retention

**Keep only as
required**

Technical Activity 3 uses synthetic or de-identified information only.



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Stakeholder sampling must represent relevant variation

Select participants deliberately across roles, experience, workload, location, access needs, and exception exposure.

PURPOSIVE SAMPLE

Choose people because their role or experience can answer a defined evidence question.

COVERAGE CHECK

Ask which perspective, shift, location, access need, or exception remains absent.

Sample size alone cannot correct a systematically missing stakeholder group.



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Classify claims before using them

Keep facts, statements, interpretations, assumptions, and decisions separate.

Class	Meaning	Example
Observed fact	Recorded event	7 of 20 forms lacked ID
Reported fact	Participant statement	Reviewer reports rework
Interpretation	Analyst explanation	Field wording may confuse
Assumption	Unverified belief	All users have stable data
Decision	Approved direction	Offline capture is in scope



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Triangulation strengthens or challenges a claim

Compare different people, methods, records, and time periods before accepting a requirement.



Conflicting evidence is a finding—not a failure

Preserve the difference, identify context, seek authoritative rules, and escalate decisions transparently.



Do not silently choose the statement that best fits the proposed solution.

Evidence is sufficient when the decision risk is acceptable

Stop gathering only when important claims have coverage, credibility, consistency, recency, and traceability.

Check	Question	Warning sign
Coverage	Are key roles and cases included?	Missing exception users
Credibility	Is the source positioned to know?	Opinion treated as fact
Consistency	Do sources agree or explain differences?	Unresolved conflict
Recency	Does evidence reflect current work?	Obsolete policy



Document the current workflow before proposing change

Capture actors, triggers, steps, decisions, handoffs, artifacts, exceptions, timing, and pain points.



Current-state documentation describes what happens now—including workarounds—not the future design.

Connect pain points to information needs and sources

This chain turns complaints into evidence questions that can later justify requirements.

Pain point	Information needed	Likely source	Possible requirement direction
Repeated returns	Which fields are incomplete?	Returned forms	Validate required fields
Unknown status	Where does work wait?	Timestamps + interviews	Track stage + owner
Inconsistent checks	Which rules differ?	Policy + reviewers	Standardize decision rules

Cybersecurity: follow evidence across trust boundaries

Case focus: incident reporting, evidence handling, validation, response support, and accountable access.

Stakeholder	Evidence to gather	Requirement direction
Reporter	Submission context + safety	Protected evidence intake
Validator	Credibility checks + exceptions	Traceable review status
Responder	Prioritization information	Actionable verified alerts
Security owner	Threats + access duties	Least privilege + audit

Web and mobile: study the complete service journey

Case focus: visitor requests, guide verification, status communication, accessibility, connectivity, and support.

Stakeholder	Evidence to gather	Requirement direction
Visitor	Search + referral needs	Clear verified information
Tourism staff	Review workflow + rules	Consistent verification
Guide/operator	Submission + correction	Transparent status
Support team	Devices + failures	Accessible resilient service

Animation and game development: creation and play

Case focus: content accuracy, asset workflow, player experience, accessibility, device limits, and facilitation.

Stakeholder	Evidence to gather	Requirement direction
Learner/player	Goals + barriers + safety	Accessible feedback
Designer/artist	Asset pipeline + versions	Reviewable production flow
Subject expert	Accuracy + approvals	Content validation
Facilitator/support	Deployment + recovery	Operable session controls

Business analytics: begin before the dashboard

Case focus: data ownership, definitions, lineage, quality, access, interpretation, and decision responsibility.

Stakeholder	Evidence to gather	Requirement direction
Data owner	Authority + sensitivity	Approved access
Data steward	Definitions + quality	Validated data rules
Analyst	Transformations + assumptions	Traceable lineage
Decision-maker	Questions + thresholds	Explainable indicators

Evidence-pack quality gate

Can another analyst trace every important claim to a protected source, understand unresolved conflicts, and see what must still be validated?



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REQUIREMENTS ENGINEERING AND VALIDATION

A requirement must remain connected to evidence

Move from verified need to specification, priority, acceptance, validation, and downstream analysis.



If the evidence changes, the requirement and every linked artifact must be reviewed.

Requirements describe different kinds of obligation

Classify each statement so missing quality, data, interface, security, privacy, and rule needs stay visible.

Type	Question answered	Example direction
Functional	What must the system do?	Record a review decision
Data + interface	What information and exchange?	Validate credential fields
Quality	How well must it perform?	Return status within target
Security + privacy	What protection is required?	Restrict evidence access
Rule + constraint	What condition or limit applies?	Retention follows policy



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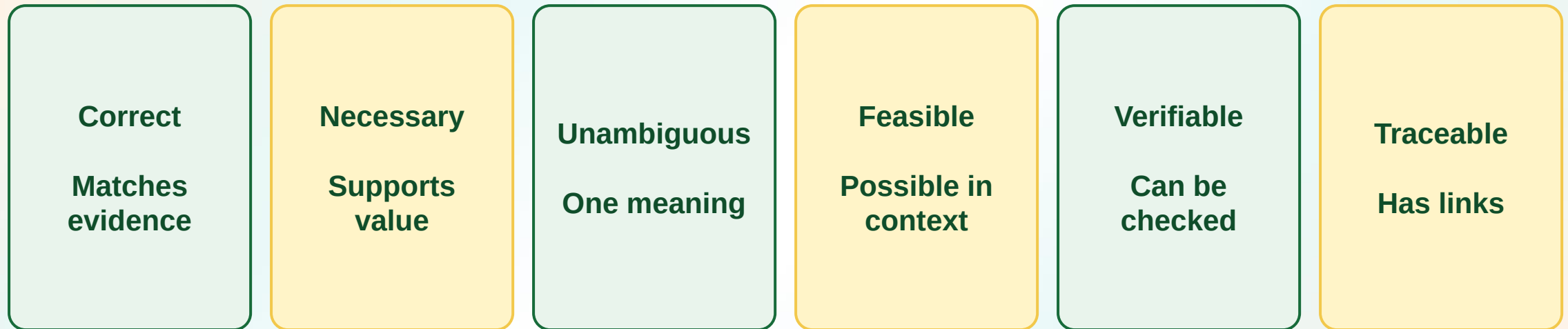


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Good requirements are clear, feasible, and testable

Check correctness, necessity, consistency, clarity, verifiability, and traceability.



A grammatically polished sentence can still be unnecessary, unsafe, or unsupported.

Replace vague adjectives with observable conditions

Rewrite ambiguous statements by naming the actor, behavior, object, condition, threshold, and verification method.

AMBIGUOUS

“The system should quickly notify authorized users about important updates.”

TESTABLE

“When a verification status changes, the system shall notify the assigned reviewer within 60 seconds and log delivery status.”

Words to challenge: easy • fast • secure • robust • user-friendly • real-time • appropriate • seamless



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A user story states value; acceptance criteria bound it

The story expresses a goal from a user perspective, while the criteria define observable completion.

As a tourism reviewer, I want incomplete guide submissions flagged before review so that avoidable returns are reduced.

Acceptance direction

- Required fields are checked before submission.
- Missing fields are identified beside the relevant entry.
- The submission is not forwarded until errors are resolved.

Choose the specification form that serves the decision

User stories, use cases, shall-statements, business rules, and constraints are complementary—not interchangeable.

Form	Best use	Core structure
User story	User value + backlog	Role • goal • benefit
Use case	Actor-system interaction	Trigger • flow • outcomes
Shall-statement	Precise obligation	Actor/system • action • condition
Business rule	Decision or policy logic	If / then or rule statement
Constraint	Mandatory boundary	Platform • law • time • scope



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Acceptance criteria must cover success and failure

Write concrete preconditions, trigger, expected behavior, limits, and exception outcomes.

Scenario	Given	When	Then
Valid submission	Required fields complete	User submits	Create record + status
Missing evidence	Credential file absent	User submits	Block + identify field
Unauthorized access	User lacks reviewer role	User opens evidence	Deny + log attempt



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MoSCoW prioritization protects the essential outcome

Classify requirements by necessity and evidence—not by who argues most strongly.

MUST

Without it, the
release fails

SHOULD

Important;
workaround exists

COULD

Useful if capacity
remains

WON'T YET

Explicitly deferred

Every **MUST** needs a defensible failure consequence and acceptance evidence.



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Value, risk, and dependencies change priority

Review conflicts, assumptions, prerequisites, and consequences before sequencing work.

Requirement	Value	Risk if absent	Dependency / conflict	Decision
Validate credentials	High	Invalid referrals	Needs rule owner	Must
Offline capture	High	Service interruption	Conflicts with live checks	Should / analyze
AI recommendations	Unclear	Bias + trust	Needs evidence + oversight	Won't yet

Use ISO/IEC 25010:2023 to check quality coverage

The product quality model provides nine characteristics for specifying, measuring, and evaluating quality.

Quality focus A	Quality focus B	Quality focus C
Functional suitability	Performance efficiency	Compatibility
Interaction capability	Reliability	Security
Maintainability	Flexibility	Safety

Select characteristics relevant to verified stakeholders and context; do not copy all nine mechanically.

Turn quality goals into measurable requirements

State the context, measure, threshold, evidence source, and acceptance method.

VAGUE QUALITY

“The system shall be secure, reliable, fast, and user-friendly.”

MEASURABLE QUALITY

“Under the agreed test load, 95% of status queries shall complete within two seconds, measured from timestamped test logs.”

Pattern: context + system behavior + threshold + measure + verification evidence



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Derive security and privacy requirements from context

Start with assets, data, threats, trust boundaries, duties, and scope before naming controls.



Example: because evidence files contain sensitive identifiers, only assigned reviewers may access them, and every access attempt must be logged.

Change control keeps the baseline trustworthy

A requirement change must show its reason, source, impact, decision, owner, version, and linked updates.



Impact review includes scope, objectives, models, data, security, privacy, feasibility, schedule, acceptance, and traceability.

Validation asks whether the right requirements are specified

Use reviews, walkthroughs, scenarios, prototypes, acceptance checks, and stakeholder confirmation.

Validation question	Method	Evidence
Correct and necessary?	Stakeholder review	Confirmed need
Complete and consistent?	Peer inspection	Defect log
Understandable?	Scenario walkthrough	Clarified wording
Testable and feasible?	Acceptance review	Measure + conditions



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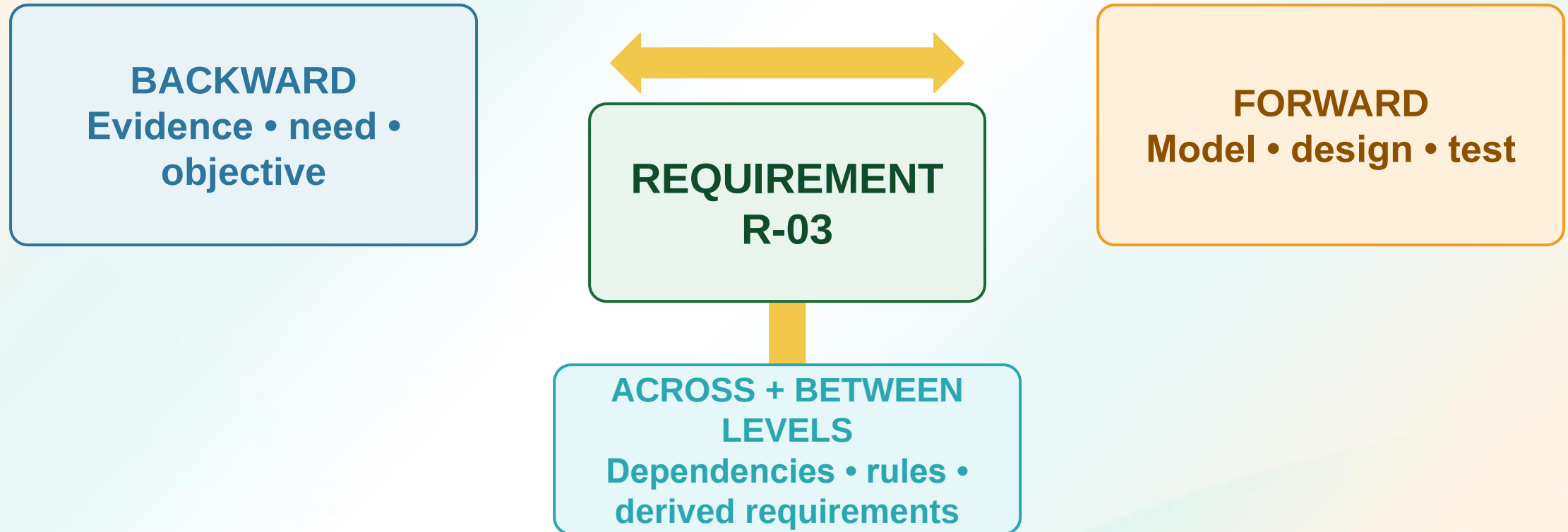
A baseline is an approved, controlled reference

Version the specification so teams know what is current, changed, and approved.

Version	State	Change	Decision evidence
0.1	Draft	Initial elicitation	Interview + records
0.2	Reviewed	Clarified exceptions	Workshop log
1.0	Baselined	Approved Module 3 set	Validation sign-off
1.1	Controlled change	Policy update	Impact + approval

Traceability must work in four directions

A requirement should be explainable backward, forward, across related items, and between levels.



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The RTM makes requirement coverage inspectable

Maintain identifiers and links from evidence through acceptance and downstream models.

ID	Source / need	Requirement	Priority	Acceptance	Model / status
R-03	INT-02 + LOG-04	Track stage + owner	Must	Status updates logged	DFD-1 / Validated
SEC-02	DATA-01 + threat	Restrict evidence access	Must	Unauthorized access denied	UC-4 / Reviewed
NFR-01	OBS-03	95% queries $\leq$ 2 s	Should	Load-test timestamps	Pending baseline

Validation is a structured conversation with evidence

Stakeholders confirm meaning and value; peers inspect quality, consistency, feasibility, acceptance, and trace links.



Silence is not approval. Record who reviewed, what was accepted, what changed, and what remains unresolved.

AI may challenge requirements—but cannot invent evidence

Use synthetic or de-identified information, validate every suggestion, revise the work, and disclose AI use.



Submit prompts and outputs, critique and validation, accepted and rejected recommendations, corrected requirements and RTM entries, and the AI Use Disclosure.

Module 3 activity path

Formative Assessment 3 [AI-Prohibited]: Complete the ungraded Canvas quiz independently after Module 3.

Class Activity 3 [AI-Integrated]: Complete the AI Stakeholder and Requirements Audit using synthetic or de-identified information, validation records, and an AI Use Disclosure.

Summative Assessment 2: Occurs after Module 4 and covers Modules 3–4.

Module 3 portfolio checkpoint

Stakeholder register and ethical evidence plan

Interview, workshop, questionnaire, observation, document, or log evidence records

Current-state workflow, pain points, information needs, and data sources

Prioritized requirements with acceptance criteria, baseline, version, and traceability

AI Stakeholder and Requirements Audit with validation and disclosure

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**VALIDATE THE REQUIREMENTS.
THEN MAKE THE SYSTEM VISIBLE.**



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